

Pre-treatment LARC tumor
(proliferative · DNA-repair-proficient · HLA-LOH± · neoantigen-presenting)

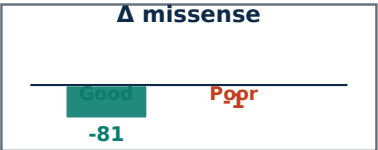


TNT
(FOLFOX / CAPOX + CRT)

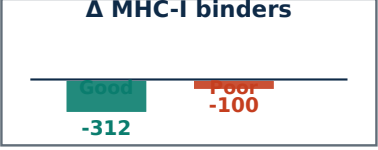
Good responders

Poor responders

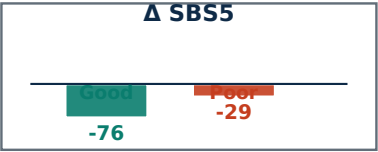
Mutation clearance



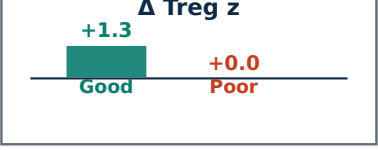
Neoantigen clone loss



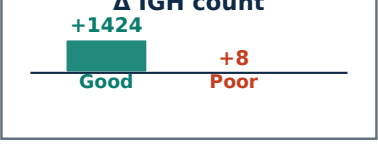
SBS5 mutation clearance



Treg / MHC-II reprogramming



BCR (IGH) infiltration



Complete / near-complete response (TRG 0-1)

Minimal molecular change

- Δ missense ≈ -1
- Δ neoantigens ≈ -100
- Δ SBS5 ≈ -29
- Δ Treg z ≈ +0.0
- Δ IGH ≈ +8

→ Primary treatment insensitivity

Partial / minimal response (TRG 2-3)